

Title

Oral pH modulation and bacterial metabolic responses following herbal rinse application: exploratory mechanistic findings

Abstract

Background: The mechanisms underlying antimicrobial efficacy of herbal oral rinses remain incompletely characterized. Secondary analysis of our randomized trial examined plaque biochemistry and bacterial metabolic markers. **Setting:** Sixty participants from the parent trial contributed biosamples. **Methods:** Plaque pH was assessed at multiple timepoints; bacterial species composition was analyzed via targeted microbiology. Participants receiving the herbal extract displayed higher oral pH readings in plaque biofilms relative to controls. Bacterial ATP production and acid generation appeared suppressed in the active-treatment group. No significant changes in other oral flora or tissue health markers were observed. The magnitude of pH change did not correlate with individual baseline caries risk. **Conclusions:** The herbal preparation may exert part of its antimicrobial effect through localized pH elevation and disruption of bacterial metabolic activity, though additional mechanisms warrant investigation.